

ProdPlan

Manufacturing Production Planning SaaS — PRD

Prepared For: Client

Budget: USD 7,500

Duration: 10 Days

Payment Structure: 3 Fixed-Price Milestones (USD 2,500 each)

Project Overview

ProdPlan is a multi-tenant SaaS platform that helps discrete manufacturers plan and track production against real constraints — customer orders, machine/work-center capacity, raw-material availability, and delivery deadlines. It replaces spreadsheet-based production planning with a single live production schedule: planners create orders, the system checks material and capacity availability, builds a day-by-day machine schedule, and gives floor supervisors and managers real-time visibility into production status against every delivery deadline.

Objectives

- Replace spreadsheet/whiteboard production planning with a single, live scheduling system.
- Automatically check machine capacity and material availability before confirming an order into the schedule.
- Give planners a visual, day-by-day machine schedule that highlights conflicts and overloads.
- Give managers real-time visibility into production status against every delivery deadline.
- Support multiple manufacturing units (tenants) on one platform, each with isolated data and configuration.
- Flag material shortages and at-risk delivery dates automatically, instead of relying on manual follow-up.

User Roles

Admin (plant/company owner — full configuration access), Production Planner (creates orders, builds and adjusts the schedule), Floor Supervisor (updates production status, views assigned schedule), and Viewer (read-only access to dashboards and reports).

Functional Requirements

- User login and signup (email/password authentication) with role-based access control.
- Multi-tenant setup so each manufacturing unit's orders, machines, materials, and schedules are fully isolated.
- Order management: create/import customer orders with product, quantity, priority, and delivery deadline.
- Machine & work-center master data: capacity per shift, working calendar, downtime/maintenance blocking.
- Material master & inventory: stock on hand, unit of measure, reorder threshold, per-product bill of materials (BOM).
- Production scheduling engine: sequences orders across machines based on capacity, material availability, and deadline priority, and flags conflicts (overloaded machine, insufficient material, missed-deadline risk).
- Visual schedule / planning board: day-by-day, machine-by-machine view (Gantt-style) with drag-to-reschedule.

- Production status tracking: stage-wise status per order (queued, in-progress, on-hold, completed) with floor-level update capability.
- Delivery deadline tracking with automatic at-risk / delayed flags visible on the dashboard.
- Admin analytics dashboard: machine utilization, on-time delivery rate, material shortages, order throughput.
- In-app and email notifications for schedule conflicts, material shortages, and approaching deadlines.
- Schedule and production report export (CSV/PDF).
- Responsive web UI for both office planning use and floor-level (tablet) status updates.

Non-Functional Requirements

- Encryption in transit and at rest.
- Per-tenant data isolation.
- Role-enforced API access.
- Full audit logging of schedule and order changes.
- Sub-2-second page loads for planning board and dashboards.
- Architecture that scales from a single pilot plant to multiple manufacturing units.
- 99.5%+ target uptime.

AI & Automation

- AI-assisted schedule optimization: given open orders, machine capacity, and material stock, suggests a conflict-minimized production sequence rather than requiring the planner to sequence manually.
- Material shortage prediction: cross-checks BOM requirements for scheduled orders against current stock and reorder lead time, and proactively flags orders at risk before they hit the floor.
- Delivery-risk detection: continuously re-evaluates the schedule as status updates come in and surfaces orders likely to miss their deadline, with the contributing reason (capacity, material, or delay upstream).
- Scheduled notification engine for conflicts, shortages, and at-risk deadlines — email and in-app.

Payment Integration

Not required for this engagement — ProdPlan is an internal operations tool, not a payments product. If a future phase adds paid subscription billing for manufacturing units, the architecture should not need to be redesigned to add it.

Project Budget

Total Project Budget: USD 7,500

Divided evenly across three milestones: Milestone 1 — USD 2,500, Milestone 2 — USD 2,500, Milestone 3 — USD 2,500.

Project Milestones

As senior full-stack engineer on this engagement, I have scoped the ten-day build into three milestones, each releasing USD 2,500 on completion:

Milestone 1	USD 2,500
• Multi-tenant architecture setup with per-tenant data isolation. • Authentication, signup/login, and role-based access control (Admin, Planner, Supervisor, Viewer). • Order management module: create, edit, and import customer orders (product, quantity, priority, delivery deadline). • Machine & work-center master data: capacity, shift calendar, downtime/maintenance windows. • Material master & inventory module with per-product bill of materials (BOM). • Base admin dashboard with order and machine list views. • Staging deployment for early client review.	

Milestone 2	USD 2,500
• Production scheduling engine: sequences orders against machine capacity and material availability, factoring in deadline priority. • Conflict detection: flags overloaded machines, insufficient material, and at-risk deadlines at scheduling time. • Visual planning board (Gantt-style, day-by-day, machine-by-machine) with drag-to-reschedule. • Production status tracking: stage-wise status per order (queued, in-progress, on-hold, completed) with floor-level update flow. • Delivery deadline tracking with automatic at-risk / delayed flags. • In-app notification framework wired to scheduling conflicts and status changes.	

Milestone 3	USD 2,500
• AI-assisted schedule optimization: conflict-minimized sequencing suggestions across open orders. • AI material shortage prediction against BOM requirements and stock levels for the live schedule. • Delivery-risk detection surfaced on the dashboard with contributing reason (capacity, material, or upstream delay). • Manager analytics dashboard: machine utilization, on-time delivery rate, material shortages, order throughput. • Email notification engine, schedule/production report export (CSV/PDF), audit logging. • Security hardening, responsive UI pass (office + floor/tablet use), and production deployment. • Full deployment & go-live: domain connection, hosting setup, SSL, production database, and final cutover (see Deployment & Go-Live section).	

Timeline

Estimated project duration: 10 days depending on feedback and revision cycles. Delivery may extend if additional features are requested mid-project.

Acceptance Criteria

All modules listed under each milestone are complete, responsive, tested, deployed to production, and delivered with source code, environment setup instructions, and a short handover/knowledge-transfer session.

Deployment & Go-Live

The following deployment tasks are included as part of Milestone 3 and completed before final handover:

- Domain connection: pointing the client's domain/subdomain to the production environment and configuring DNS records (A/CNAME).
- Hosting setup: provisioning and configuring the production hosting environment (e.g., Hostinger/VPS or equivalent cloud host).
- SSL certificate installation for secure (HTTPS) access.
- Production database setup, connection security, and backup configuration.
- Environment variables and secrets configured for the live environment (separate from staging).
- Final smoke testing on the live domain across all core workflows.
- Go-live cutover and post-deployment verification.

Future Scope

- Native mobile apps (iOS/Android) for floor supervisors.
- Machine IoT/sensor integration for automatic status and downtime capture.
- ERP integration (e.g., Tally, SAP, Zoho) for order and inventory sync.
- Multi-plant, multi-site scheduling with cross-site capacity balancing.
- What-if scenario planning (simulate schedule impact of a new rush order).
- Subscription/billing management for the SaaS itself.